

E2EE Chat App

*End-to-End Encrypted Chat Application
Using Post-Quantum Cryptography*

Scalable Algorithms for Data Science Laboratory



Q3-Chat

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I. Introduction

Much like Y2K hung over the head of 1990s programmers, quantum computation is now hanging over the head of modern programmers. Once quantum computation has become common, messages sent with traditional cryptography will become trivially accessible. If the cryptography community waits until the last possible moment to switch to post-quantum cryptography, then there will still be plenty of previously sent messages to decrypt. Secure communications must begin preparing for quantum computing now.

In addition to post-quantum encryption, modern chat applications should ensure security through end-to-end encryption and decentralization. End-to-end encryption ensures that only the sender and receiver can access the messages. Decentralization ensures that no single authority, government, or emergency can realistically prevent server access.

II. Background and Related Work

Post-quantum cryptography (PQC) focuses on developing cryptographic algorithms that remain secure against both classical and quantum computers. Most encrypted communication today uses a combination of public-key and symmetric-key cryptography. Public-key cryptography allows two parties to establish a shared secret while communicating over an untrusted network. Once this secret has been established, a symmetric encryption algorithm such as the Advanced Encryption Standard (AES) can efficiently encrypt the actual conversation.

A major concern with this approach is that many widely used public-key algorithms are not expected to remain secure against sufficiently powerful quantum computers. RSA and elliptic-curve cryptography rely on mathematical problems that are extremely difficult for conventional computers to solve. However, Shor's algorithm would allow a sufficiently powerful quantum computer to solve these problems much more efficiently. Symmetric encryption is affected differently and can remain useful with sufficiently large key sizes, making algorithms such as AES-256 still applicable to post-quantum systems.

Although cryptographically relevant to quantum computers do not currently exist, encrypted information may already be at risk through an attack known as "harvest now, decrypt later." An attacker can collect and store encrypted network traffic today without being able to decrypt it. If sufficiently powerful quantum computers become available in the future, the attacker could potentially recover information that was protected using vulnerable public-key cryptography. This is particularly concerning information that needs to remain confidential for many years. Because of this threat, organizations are beginning to transition toward quantum-resistant cryptography before large-scale quantum computers become available.

NIST Post-Quantum Cryptography Standards

To prepare for this transition, the U.S. National Institute of Standards and Technology (NIST) began its Post-Quantum Cryptography Standardization Project in 2016. After several rounds of public evaluation, NIST published its first three finalized post-quantum cryptography standards in August 2024: FIPS 203, FIPS 204, and FIPS 205.

FIPS 203 defines the Module-Lattice-Based Key-Encapsulation Mechanism (ML-KEM), which was developed from CRYSTALS-Kyber. ML-KEM allows two parties to communicate over an untrusted network to establish shared secret material. That secret can then be used to derive symmetric keys for encrypting their communication. This makes ML-KEM particularly relevant to Q3-Chat because two clients need to establish encryption keys without exposing those keys to the servers facilitating their connection.

NIST also standardized two digital signature algorithms. FIPS 204 defines ML-DSA, developed from CRYSTALS-Dilithium, while FIPS 205 defines SLH-DSA, developed from SPHINCS+. These algorithms serve a different purpose from ML-KEM. A key encapsulation mechanism can establish shared secret information, while digital signatures can verify the authenticity and integrity of information. Together, these standards provide important building blocks for replacing cryptographic systems that may become vulnerable to quantum computers.

End-to-End Encryption and Related Work

Post-quantum cryptography alone does not guarantee private communication. Q3-Chat must also use end-to-end encryption (E2EE), where a message is encrypted on the sender's device and is not decrypted until it reaches the intended recipient. Servers may transport or store messages, but they should never have the private keys needed to read them.

This separation is important to Q3-Chat's architecture. A server may be responsible for user registration, locating users, distributing public cryptographic information, forwarding messages, and storing encrypted messages for offline clients. However, encryption and decryption occur entirely on client devices. Therefore, even if an attacker gains access to a Q3-Chat server, the attacker should not immediately gain access to the plaintext contents of stored conversations.

Existing encrypted messaging systems provide useful related work for this approach. Signal, for example, introduced its Post-Quantum Extended Diffie-Hellman protocol (PQXDH) in 2023. PQXDH combines traditional elliptic-curve cryptography with post-quantum cryptography when establishing shared secrets between users. Signal's approach demonstrates how post-quantum protection can be integrated into an existing end-to-end encrypted messaging system.

PQXDH is particularly relevant because it supports asynchronous communication. A recipient does not have to be online when another user begins a conversation. Public cryptographic information can instead be stored on a server and retrieved by another client when establishing an encrypted session. The server facilitates the exchange but does not obtain the resulting shared secret. This is like Q3-Chat's intended architecture, where servers help users establish connections and route messages without decrypting.

Modern encrypted messaging systems also use techniques such as forward secrecy to limit the damage caused by compromised keys. Signal's Double Ratchet algorithm, for example, continuously derives new encryption keys as messages are exchanged. Older keys can then be deleted, preventing a current key from automatically providing access to the entire previous conversation. This demonstrates that secure messaging involves more than choosing a strong

encryption algorithm. Key establishment, authentication, key rotation, key storage, and device management must also be considered.

Decentralization

Q3-Chat will also use a decentralized server architecture. Typical messaging applications rely on infrastructure operated by a single organization. A decentralized system would allow multiple servers to participate in the network. For example, User A could be connected to Server A while User B is connected to Server B. The servers would communicate with each other to locate users and route encrypted messages, while encryption and decryption would remain entirely on the client devices.

This approach separates message transportation from message confidentiality. Servers need enough information to determine where encrypted messages should be delivered, but they do not need the cryptographic keys required to read them. It also reduces the application's dependence on a single server. However, decentralization creates additional challenges, including authenticating communication between servers, securely distributing users' public keys, synchronizing encrypted messages across devices, and preventing a malicious server from substituting cryptographic information.

These challenges will be important considerations during the development of Q3-Chat. By combining decentralized communication with end-to-end encryption and standardized post-quantum cryptography, the project will explore a messaging architecture designed to protect user communications from both current security threats and potential future attacks involving quantum computers.

III. Project Overview

The primary objective of our project is to produce an end-to-end encrypted chat application that uses post-quantum cryptography. Post-quantum cryptography enables our application to remain secure both today and when quantum computing has broken standard encryption schemes. Security and anonymity are paramount to our application. For our application to attract potential users, it must be auditable and anonymously accessible.

The chat application requires a server for hosting. These servers will be responsible for handling user registration and initiating connections. Once a server has initiated a connection, its sole responsibility will be forwarding messages between servers, serving encrypted messages to clients, and storing encrypted message history. The message history exists solely to backup messages for clients changing devices. At no point will the server be able to decrypt stored messages.

To ensure redundancy and resistance to outside forces, our servers will also be decentralized. Decentralization allows for client-side applications to choose to register and communicate with any one of multiple independent servers. This server will then be able to forward messages to clients communicating on other servers. The client-side application will be responsible for all message decryptions. It will be the sole storage location for the encryption keys. Device management and device switching will be handled by and between the client devices.

IV. Client and Stakeholder Identification and Preferences

Our primary client is Dr. James Halvorsen, a liaison for the Scalable Algorithms for Data Science Laboratory (SCADS), led by Dr. Assefaw Gebremedhin. Dr. Parteek Kumar, associate professor at the School of Electrical Engineering & Computer Science, serves as a faculty mentor. Collaboration will occur through weekly hybrid meetings.

The main stakeholder is the general public, particularly those interested in private applications and lifelong protection of chat threads. Another stakeholder is the industrial complex. Included are healthcare companies who require patient-provider confidentiality, the defense sector which relies on mission-critical infrastructure, and financial institutions that must protect client assets.

Regarding client preferences, the E2EE application will support cross-platform compatibility, low latency, and various message mediums such as text, documents, and video. The application should be decentralized to maximize the number of users as well as implement NIST-verified post-quantum algorithms available in libraries like OpenSSL. Additionally, messages must not be decryptable by anyone other than the sender and receiver. To ensure this objective is met, a security audit will be conducted on the final product.

V. Team Member Introduction

Name: James Richards-Perhatch

Major: Computer Science



I am a computer science major at Washington State University with a focus on cybersecurity. I am a member of the WSU Cybersecurity Group as well as both an SFS and VICEROY scholar. My course work has given me experience in computer architecture, software engineering, cryptography, and network security.

Name: Mitchell Milander

Major: Computer Science



Hello, I am a computer science student at Washington State University with strong foundations in software development, cybersecurity, and advanced mathematics. I have experience in research, hands-on technical projects, and interdisciplinary innovation through academic programs and internships. It is my passion to build and improve technology through analytical thinking and programming.

Name: Aiden Vezzetti

Major: Computer Science



Hello, I am a senior computer science student minoring in mathematics at Washington State University with an emphasis on cybersecurity. I have experience with software development, networking, reverse engineering, and cybersecurity through coursework, and personal projects.

VI. Glossary

E2EE: "A secure communication process that encrypts data before transferring it to another endpoint" (IBM).

Post-Quantum Cryptography: "Mathematical algorithms designed to be secure against the computational capabilities of a cryptographically relevant quantum computer" (Palo Alto Networks).

VII. References

IBM, "What Is End-to-End Encryption?" <https://www.ibm.com/think/topics/end-to-end-encryption>

Palo Alto Networks, "What is Post-Quantum Cryptography (PQC)? A Complete Guide"
<https://www.paloaltonetworks.com/cyberpedia/what-is-post-quantum-cryptography-pqc>